

# BUSINESS REQUIREMENTS DOCUMENT (BRD)

---

## Project Information

|                        |                                        |
|------------------------|----------------------------------------|
| <b>Project Name</b>    | Coffee Shop Profit Analysis            |
| <b>Project Type</b>    | Data Analytics / Business Intelligence |
| <b>Domain</b>          | Retail / Food & Beverage               |
| <b>Prepared By</b>     | Your Team Members                      |
| <b>Version</b>         | 1.0                                    |
| <b>Document Status</b> | Final                                  |
| <b>Date</b>            | DD/MM/YYYY                             |

---

## Table of Contents

1. Executive Summary
2. Business Background
3. Problem Statement
4. Business Objectives
5. Project Scope
  6. 5.1 In Scope
  7. 5.2 Out of Scope
8. Business Requirements
9. Functional Requirements
10. Business Questions
11. Data Analysis Requirements

12. Key Performance Indicators (KPIs)

13. Assumptions

14. Dashboard Requirements

15. Expected Deliverables

16. Success Criteria

17. Business Recommendations

18. Future Enhancements

19. Approval

Appendix A – Glossary

Appendix B – Abbreviations

Appendix C – References

---

# 1. Executive Summary

The purpose of this project is to analyze the Coffee Shop Sales dataset and identify business opportunities that can increase profitability. Although the coffee shop generates consistent daily sales, management lacks sufficient analytical insights to understand which products, categories, and business hours contribute the most to overall revenue and profit.

This project applies Data Analytics and Business Intelligence techniques to analyze historical sales data, evaluate customer purchasing behavior, identify business trends, and monitor key business metrics. The analysis will help management make informed decisions regarding product performance, sales optimization, and operational improvements.

The final outcome of the project will include a cleaned dataset, exploratory data analysis (EDA), an interactive Power BI dashboard, and actionable business recommendations to improve profitability and support strategic decision-making.

---

## 2. Business Background

ABC Coffee Shop operates in the Food and Beverage industry and serves a variety of coffee beverages, bakery items, snacks, and other food products. The business records hundreds of customer transactions daily, generating valuable operational data.

Currently, management relies primarily on manual reports and basic sales summaries to evaluate business performance. While these reports provide overall sales figures, they do not explain the underlying business trends, customer purchasing behavior, or product performance.

As competition in the retail coffee industry continues to increase, data-driven decision-making has become essential. By analyzing historical sales data, the business can identify profitable products, understand customer buying patterns, optimize operational performance, and improve overall profitability.

---

## 3. Problem Statement

Although the coffee shop generates regular sales, overall profitability remains inconsistent. Management lacks a centralized analytical solution that provides detailed insights into product performance, customer purchasing behavior, and sales trends.

As a result, management is unable to identify:

- High-performing products
- Low-performing products
- Peak sales periods
- Customer purchasing patterns
- Business growth opportunities

To address these challenges, the organization requires a Business Intelligence solution capable of transforming raw sales data into meaningful business insights.

---

## 4. Business Objectives

The primary objectives of this project are:

- Increase overall business profitability.
- Identify the highest-performing products.
- Identify products requiring improvement.
- Analyze customer purchasing behavior.
- Monitor business performance using Key Performance Indicators (KPIs).
- Support data-driven decision-making.
- Develop an interactive Power BI dashboard.

- Provide actionable recommendations for business growth.
- 

## 5. Project Scope

### 5.1 In Scope

The following activities are included within the scope of this project:

- Dataset Understanding
  - Data Cleaning
  - Data Validation
  - Exploratory Data Analysis (EDA)
  - Product Performance Analysis
  - Sales Performance Analysis
  - Category Performance Analysis
  - Time-Based Sales Analysis
  - KPI Calculation
  - Dashboard Development using Power BI
  - Business Insights
  - Business Recommendations
- 

### 5.2 Out of Scope

The following activities are outside the scope of this project:

- Mobile Application Development
  - Website Development
  - Online Ordering System
  - Customer Relationship Management (CRM)
  - Inventory Management System
  - Marketing Campaign Execution
  - Employee Management System
- 

## 6. Business Requirements

The proposed solution should enable the coffee shop management to monitor business performance, identify growth opportunities, and make strategic business decisions based on analytical insights.

| BR ID | Business Requirement               | Priority |
|-------|------------------------------------|----------|
| BR-01 | Analyze overall sales performance. | High     |

---

| BR ID | Business Requirement                                            | Priority |
|-------|-----------------------------------------------------------------|----------|
| BR-02 | Identify high-performing products.                              | High     |
| BR-03 | Identify low-performing products.                               | High     |
| BR-04 | Analyze sales trends by hour, day, and month.                   | High     |
| BR-05 | Evaluate category-wise sales performance.                       | Medium   |
| BR-06 | Generate meaningful business insights.                          | High     |
| BR-07 | Support management decision-making through Power BI dashboards. | High     |
| BR-08 | Recommend strategies to improve profitability.                  | High     |

## 7. Functional Requirements

The solution should provide the following functional capabilities.

| FR ID | Functional Requirement                                                                           | Priority |
|-------|--------------------------------------------------------------------------------------------------|----------|
| FR-01 | Import the Coffee Shop Sales dataset.                                                            | High     |
| FR-02 | Validate and clean the dataset by handling missing values, duplicates, and inconsistent records. | High     |
| FR-03 | Perform Exploratory Data Analysis (EDA).                                                         | High     |
| FR-04 | Analyze product-wise sales performance.                                                          | High     |
| FR-05 | Analyze category-wise sales performance.                                                         | High     |
| FR-06 | Analyze daily, weekly, monthly, and hourly sales trends.                                         | High     |
| FR-07 | Calculate Key Performance Indicators (KPIs).                                                     | High     |
| FR-08 | Develop an interactive Power BI dashboard.                                                       | High     |
| FR-09 | Generate business insights based on analytical findings.                                         | High     |
| FR-10 | Provide business recommendations to improve profitability.                                       | High     |

## 8. Business Questions

The following business questions will guide the analysis and help management make informed business decisions.

| Question ID | Business Question                                                 | Business Purpose                                        |
|-------------|-------------------------------------------------------------------|---------------------------------------------------------|
| BQ-01       | Which product generates the highest revenue?                      | Identify top-performing products.                       |
| BQ-02       | Which product has the lowest sales?                               | Identify products requiring improvement or replacement. |
| BQ-03       | Which product category contributes the most to total revenue?     | Understand category performance.                        |
| BQ-04       | During which hours does the coffee shop experience maximum sales? | Optimize staffing and operational planning.             |
| BQ-05       | Which day of the week generates the highest revenue?              | Plan promotional campaigns and staffing.                |
| BQ-06       | Which month records the highest sales?                            | Identify seasonal sales trends.                         |
| BQ-07       | What is the average order value?                                  | Understand customer spending behavior.                  |
| BQ-08       | Which products should be promoted to maximize profitability?      | Improve sales and marketing strategies.                 |

## 9. Data Analysis Requirements

To achieve the project objectives, the following analyses will be performed.

| Analysis                     | Objective                               | Expected Outcome                             |
|------------------------------|-----------------------------------------|----------------------------------------------|
| Sales Analysis               | Analyze overall sales performance.      | Understand business growth.                  |
| Revenue Analysis             | Evaluate total revenue generated.       | Measure financial performance.               |
| Product Performance Analysis | Compare product-wise sales and revenue. | Identify best and worst performing products. |
| Category Analysis            | Analyze category contribution.          | Understand category profitability.           |

| Analysis                   | Objective                          | Expected Outcome              |
|----------------------------|------------------------------------|-------------------------------|
| Hourly Sales Analysis      | Evaluate sales by hour.            | Identify peak business hours. |
| Daily Sales Analysis       | Compare daily sales performance.   | Understand weekday trends.    |
| Monthly Sales Analysis     | Analyze monthly revenue trends.    | Identify seasonal demand.     |
| Customer Purchase Analysis | Study purchasing patterns.         | Improve customer experience.  |
| Trend Analysis             | Analyze business growth over time. | Support strategic planning.   |

## 10. Key Performance Indicators (KPIs)

The following KPIs will be used to measure business performance and evaluate project success.

| KPI                    | Description                                        | Business Value                              |
|------------------------|----------------------------------------------------|---------------------------------------------|
| Total Sales            | Total sales generated by the coffee shop.          | Measures overall business performance.      |
| Total Revenue          | Total revenue earned from all transactions.        | Indicates financial growth.                 |
| Total Orders           | Total number of completed customer orders.         | Measures customer activity.                 |
| Average Order Value    | Average amount spent per customer transaction.     | Evaluates customer spending behavior.       |
| Best Selling Product   | Product with the highest sales volume.             | Supports inventory and marketing decisions. |
| Lowest Selling Product | Product with the lowest sales volume.              | Identifies products requiring improvement.  |
| Sales by Category      | Revenue contribution of each product category.     | Helps evaluate category performance.        |
| Sales by Hour          | Revenue generated during different business hours. | Supports workforce planning.                |
| Sales by Day           | Revenue generated on each day of the week.         | Identifies peak business days.              |
| Monthly Revenue        | Revenue generated each month.                      | Measures seasonal business performance.     |

| KPI            | Description                                           | Business Value                    |
|----------------|-------------------------------------------------------|-----------------------------------|
| Revenue Growth | Percentage increase or decrease in revenue over time. | Evaluates business growth trends. |

## 11. Assumptions

The following assumptions are considered during project execution.

| Assumption ID | Assumption                                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------|
| A-01          | The Coffee Shop Sales dataset is complete and accurately represents business transactions.                 |
| A-02          | Product names and categories are correctly maintained.                                                     |
| A-03          | Historical sales data reflects actual business performance.                                                |
| A-04          | There are no major operational changes during the analysis period.                                         |
| A-05          | Data cleaning activities will resolve missing or inconsistent records without affecting business insights. |

## 12. Dashboard Requirements

The Power BI dashboard should provide an interactive and user-friendly interface that enables management to monitor business performance efficiently.

| Dashboard Component | Purpose                                                        |
|---------------------|----------------------------------------------------------------|
| KPI Cards           | Display Total Sales, Revenue, Orders, and Average Order Value. |
| Line Chart          | Visualize monthly sales and revenue trends.                    |
| Clustered Bar Chart | Compare product-wise sales performance.                        |
| Column Chart        | Compare category-wise sales performance.                       |
| Pie / Donut Chart   | Display category contribution to total revenue.                |
| Hourly Sales Chart  | Analyze sales distribution across different business hours.    |
| Daily Sales Chart   | Compare sales across weekdays.                                 |
| Top 10 Products     | Highlight highest-performing products.                         |



| Dashboard Component | Purpose                                                       |
|---------------------|---------------------------------------------------------------|
| Bottom 10 Products  | Identify products requiring improvement.                      |
| Slicers             | Enable filtering by Date, Product Category, and Product Name. |

## 13. Expected Deliverables

Upon successful completion of the project, the following deliverables will be provided.

| Deliverable                            | Description                                            |
|----------------------------------------|--------------------------------------------------------|
| Cleaned Dataset                        | Dataset after data cleaning and validation.            |
| Exploratory Data Analysis (EDA) Report | Summary of data exploration and key findings.          |
| Interactive Power BI Dashboard         | Dashboard containing KPIs and business visualizations. |
| KPI Report                             | Summary of business performance indicators.            |
| Business Insights Report               | Analytical findings and observations.                  |
| Business Recommendations               | Actionable recommendations to improve profitability.   |
| Final Project Documentation            | Complete project documentation, including the BRD.     |

## 14. Success Criteria

The project will be considered successful if the following objectives are achieved.

| Success Criteria      | Expected Outcome                                                         |
|-----------------------|--------------------------------------------------------------------------|
| Data Quality          | Dataset is cleaned, validated, and suitable for analysis.                |
| Accurate Analysis     | Business insights are based on reliable and accurate data.               |
| Dashboard Performance | Dashboard is interactive, responsive, and easy to use.                   |
| KPI Availability      | All required KPIs are calculated correctly.                              |
| Business Insights     | High-performing and low-performing products are successfully identified. |
| Decision Support      | Dashboard supports effective management decision-making.                 |
| Business Value        | Recommendations help improve profitability and operational efficiency.   |

## 15. Business Recommendations

Based on the analysis of the Coffee Shop Sales dataset, the following recommendations are proposed to improve profitability and overall business performance.

| Recommendation ID | Recommendation                                                                 | Expected Business Impact                                   | Priority |
|-------------------|--------------------------------------------------------------------------------|------------------------------------------------------------|----------|
| REC-01            | Promote high-selling products through combo offers and seasonal discounts.     | Increase sales and revenue.                                | High     |
| REC-02            | Review the pricing strategy of low-performing products.                        | Improve product profitability.                             | High     |
| REC-03            | Introduce promotional offers during low-sales hours to attract more customers. | Increase customer traffic during off-peak hours.           | High     |
| REC-04            | Maintain adequate inventory for high-demand products.                          | Prevent stock shortages and improve customer satisfaction. | Medium   |
| REC-05            | Schedule additional staff during peak business hours.                          | Improve operational efficiency and customer service.       | Medium   |
| REC-06            | Launch a customer loyalty program to encourage repeat purchases.               | Increase customer retention and long-term revenue.         | Medium   |
| REC-07            | Regularly monitor KPIs using the Power BI dashboard.                           | Enable data-driven business decisions.                     | High     |

## 16. Future Enhancements

The current project focuses on descriptive analytics. Future enhancements can extend the solution by incorporating predictive and advanced analytics.

| Enhancement       | Description                                 | Business Benefit                         |
|-------------------|---------------------------------------------|------------------------------------------|
| Sales Forecasting | Predict future sales using historical data. | Improve business planning and budgeting. |
| Demand Prediction | Forecast customer demand for products.      | Optimize inventory management.           |

| Enhancement             | Description                                     | Business Benefit                      |
|-------------------------|-------------------------------------------------|---------------------------------------|
| Customer Segmentation   | Group customers based on purchasing behavior.   | Enable targeted marketing campaigns.  |
| Recommendation System   | Recommend products based on previous purchases. | Increase average order value.         |
| Inventory Optimization  | Optimize stock levels using sales trends.       | Reduce wastage and stock shortages.   |
| Real-Time Dashboard     | Connect Power BI with live sales data.          | Enable real-time business monitoring. |
| Profit Prediction Model | Use machine learning to predict future profit.  | Support strategic decision-making.    |

## 17. Approval

This Business Requirements Document (BRD) has been reviewed and approved for project execution.

| Name              | Role           | Signature | Date |
|-------------------|----------------|-----------|------|
| Your Team Members | Project Team   | ___       | ___  |
| Project Mentor    | Faculty Mentor | ___       | ___  |
| Project Guide     | Academic Guide | ___       | ___  |

## Appendix A – Glossary

| Term     | Definition                                        |
|----------|---------------------------------------------------|
| BRD      | Business Requirements Document                    |
| KPI      | Key Performance Indicator                         |
| EDA      | Exploratory Data Analysis                         |
| BI       | Business Intelligence                             |
| CSV      | Comma-Separated Values                            |
| Power BI | Business Intelligence and Data Visualization Tool |

## Appendix B – Abbreviations

| Abbreviation | Full Form                 |
|--------------|---------------------------|
| BR           | Business Requirement      |
| FR           | Functional Requirement    |
| KPI          | Key Performance Indicator |
| EDA          | Exploratory Data Analysis |
| BI           | Business Intelligence     |
| CSV          | Comma-Separated Values    |

## Appendix C – References

The project is based on the following resources:

- Coffee Shop Sales Dataset
- Microsoft Power BI Documentation
- Business Intelligence Best Practices
- Data Analytics Methodology
- Project Requirement Notes

## Document Conclusion

This Business Requirements Document (BRD) provides a structured overview of the Coffee Shop Profit Analysis project. It defines the business problem, project objectives, business and functional requirements, analytical requirements, dashboard expectations, and strategic recommendations.

The project leverages data analytics and Power BI to transform historical sales data into meaningful business insights. By analyzing sales trends, product performance, customer purchasing behavior, and key performance indicators, the solution enables management to make informed, data-driven decisions.

The interactive dashboard and recommendations developed through this project will help the coffee shop optimize operations, improve product performance, enhance customer satisfaction, and increase overall profitability.

This BRD serves as the primary reference document for planning, executing, and evaluating the Coffee Shop Profit Analysis project.